



Blood Donation Management System Software Requirement Specification (SRS)

By: Sohaila ElSayed Ibrahim
Eman Sabry Badawy
Sara Mohamed Hamza

Under Supervision of: Dr. Ahmed Sedik –
Eng. Heidi Ahmed
Biomedical Engineering Year (3) – 2nd Semester
2025 - 2026

1. Introduction

1.1 Purpose

The purpose of this document is to provide a complete description of the functional and non-functional requirements for the **Online Blood Donation Management System**. This system aims to create an efficient, reliable, and user-friendly platform that connects blood donors with hospitals in urgent need of blood transfusions in Egypt.

The system will reduce the time and effort required for blood donation by allowing donors to register via a simple mobile/web interface and receive notifications when nearby hospitals need their specific blood type. It will also help hospitals find the nearest compatible donors quickly, minimize manual paperwork, and improve blood availability during emergencies, accidents, surgeries, pregnancy complications, and cancer treatments.

1.2 Scope

The proposed system includes:

- Web-based application for hospitals and administrators.
- Centralized MySQL database to store donor information, hospital details, and blood requests.
- Real-time notification system (using Firebase Cloud Messaging).
- Location-based search for nearest donors.
- Admin (customer service) module to manage users, monitor stock, and send confirmations/reminders.

Out of Scope:

- mobile application (can be added in future).
- Physical blood storage and lab testing management.
- Payment gateway for blood units.

1.3 Definitions, Acronyms, and Abbreviations

- **Donor:** General public who wishes to donate blood.
 - **Hospital:** Authorized medical facility requesting blood.
 - **Admin/Customer Service:** System administrator responsible for oversight.
 - **FCM:** Firebase Cloud Messaging.
 - **Haversine:** Formula used to calculate distance between two geographic points.
-

2. Overall Description

2.1 Product Perspective Currently, blood banks in Egypt are either independent or limited to specific hospital networks. This leads to delays in finding donors, especially for rare blood types, and forces hospitals to ask patients' families to donate or pay for blood. The proposed system provides a unified platform that connects donors directly with nearby hospitals in real time, making the process faster, safer, and more accessible.

2.2 Product Functions

- Donor registration and profile management.
- Hospital registration and blood request creation.
- Automatic matching and notification of nearest compatible donors.
- Admin management of users and requests.
- Post-donation confirmation and 3-month reminder.
- Information screen about blood donation rules.
- Smart Compatibility Engine: Utilizing a blood compatibility matrix to match donors and recipients beyond identical types.
- Automated Deferral Logic: System-enforced 3-month waiting period and medical rejection based on physiological screening results.

2.3 User Classes and Characteristics

1. **Blood Donors (General Public):** Age 18-60, basic smartphone literacy, want simple and fast registration.
2. **Hospitals:** Authorized medical staff, need quick access to nearby donors.
3. **System Administrators (Customer Service):** University/hospital staff, responsible for monitoring and support.

2.4 Operating Environment

- **Web:** React+ MySQL + node.js
- **Database:** MySQL.
- **Notifications:** Firebase Cloud Messaging.
- **Location:** GPS on donor's mobile.

2.5 Design and Implementation Constraints

- Use node.js + MySQL for backend.
- Passwords must be hashed (not stored in plain text).

2.6 Assumptions and Dependencies

- Users have internet access.
 - Hospitals are pre-approved by admin.
 - Donors provide accurate location and contact information.
 - Google Maps/Geocoding API (optional for better location accuracy).
-

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Donor Module (Web)

- FR-D1: Donor shall be able to register with name, age, phone, address, blood type, gender, and location, donation count, weight, Height, last donation date, diagnosis, image.
- FR-D2: Donor shall login using username and password.
- FR-D3: Donor shall update his/her profile (including availability and last donation date).
- FR-D4: Donor shall receive push notification when a nearby hospital needs his/her blood type.
- FR-D5: Donor shall view details of the blood request (hospital name, location, contact).
- FR-D6: Donor shall view information page about donation rules and eligibility.

3.2.2 Hospital Module (Web)

- FR-H1: Hospital shall register with name, phone, address, and location, license number.
- FR-H2: Hospital shall login.
- FR-H3: Hospital shall create a blood request (blood type needed, quantity, urgency).
- FR-H4: Hospital shall view list of nearby compatible donors with distance.
- FR-H5: Hospital shall contact donors directly via the displayed phone number.

3.2.3 Admin Module (Web)

- FR-A2: Admin shall register .
- FR-A2: Admin shall login special credentials .
- FR-A3: Admin shall approve/reject hospital registrations.
- FR-A4: Admin shall add/delete donors or hospitals.
- FR-A5: Admin shall monitor all blood requests and their status.
- FR-A6: Admin shall send confirmation message to donor after donation.
- FR-A7: Admin shall send reminder to donor after 3 months from last donation.

- FR-A8: Admin shall generate reports (number of donors, requests, blood types availability).

3.2.4 System Features

- Automatic matching of blood type and location (using Haversine formula).
- Notification system for urgent requests.
- Security: Password not visible during registration (masked input).

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

- System shall respond to requests within 5 seconds.
- Notification shall be delivered within 30 seconds of request creation.

3.3.2 Security Requirements

- All passwords shall be stored using strong hashing (bcrypt/PHP password_hash).
- Role-based access control (Donor / Hospital / Admin).
- Protection against SQL injection and XSS.
- "**Medical Data Privacy:** All clinical screening results and health history must be encrypted. Access is strictly restricted to authorized medical personnel to ensure patient confidentiality (HIPAA-compliant principles)."

3.3.3 Reliability and Availability

- System shall be available 24/7.
- Database backup shall be performed daily.

3.3.4 Usability

- Interface shall be simple and easy to use for all age groups.
- system shall have intuitive navigation.

3.3.5 Scalability

- System shall support adding mobile app and cloud hosting in future.